World Facts Handbook - Chapter 1: Physical Geography

Mount Everest is Earth's highest mountain above sea level, rising to 8,849 metres (29,032 feet) on the border between Nepal and the Tibet Autonomous Region of China. The mountain is part of the Himalayan range, which formed when the Indian Plate collided with the Eurasian Plate roughly fifty million years ago. The summit sits in the so-called death zone, where atmospheric pressure is about one third of sea level and oxygen availability is insufficient to sustain human life for long periods. The first confirmed ascent was made on 29 May 1953 by Edmund Hillary of New Zealand and Tenzing Norgay, a Sherpa of Nepal.

The Mariana Trench is the deepest oceanic trench on Earth, located in the western Pacific Ocean about 200 kilometres east of the Mariana Islands. Its deepest point, Challenger Deep, lies at approximately 10,935 metres (35,876 feet) below sea level. At that depth the pressure exceeds 1,000 times standard atmospheric pressure and sunlight never penetrates. In 1960 Jacques Piccard and Don Walsh reached the bottom in the bathyscaphe Trieste, and in 2012 the filmmaker James Cameron completed a solo descent.

The Sahara is the largest hot desert in the world, covering roughly 9.2 million square kilometres across North Africa, comparable in size to the entire United States. Despite its reputation, only about one quarter of the Sahara is sand; most of it is rocky plateau, gravel plain and dry valley. Its highest point is Emi Koussi, a volcano in Chad reaching 3,415 metres, and the Nile, the longest river in Africa, crosses the desert on its way to the Mediterranean Sea.

Chapter 2: Oceans, Rivers and Water Systems

The Pacific Ocean is the largest and deepest of Earth's five oceans, covering about 165 million square kilometres, an area greater than all of the planet's land combined. It contains the Mariana Trench and the Ring of Fire, a horseshoe-shaped belt of about 450 volcanoes and frequent earthquakes that traces the boundaries of several tectonic plates around the basin. Ocean currents such as the Kuroshio and the Humboldt Current redistribute heat between the tropics and the poles and strongly influence regional climate.

The Amazon is the largest river in the world by discharge, releasing an average of about 209,000 cubic metres of water per second into the Atlantic Ocean, roughly twenty percent of all river water entering the world's oceans. Its drainage basin covers around 7 million square kilometres, most of it in Brazil, and the river has more than 1,100 tributaries. The Amazon rainforest, sometimes called the lungs of the Earth, produces an estimated twenty percent of the oxygen generated by terrestrial plants and hosts about ten percent of all known species.

Lake Baikal in southern Siberia is the world's deepest and oldest freshwater lake, reaching a maximum depth of 1,642 metres and estimated to be 25 million years old. It holds roughly 23,600 cubic kilometres of fresh water, about twenty percent of the world's unfrozen surface fresh water, more than all of the North American Great Lakes combined. Baikal is also famous for its exceptional transparency and for the nerpa, the only exclusively freshwater seal species.

Chapter 3: Human Geography and Demography

Tokyo is the world's largest metropolitan area, with a population of roughly 37 million people in the greater Tokyo region, surpassing the next largest agglomerations, Delhi and Shanghai. The city became Japan's capital in 1868, when the emperor moved from Kyoto, and it is the seat of the country's government, financial markets and largest universities. Japan itself has one of the oldest populations on Earth, with about twenty-nine percent of its citizens aged 65 or older.

Africa is the second largest continent by both land area and population, home to more than 1.4 billion people across 54 recognised sovereign states. Nigeria is the most populous country on the continent, with an estimated 223 million inhabitants, followed by Ethiopia and Egypt. Africa also has the youngest population of any continent: the median age is about nineteen years, and the United Nations projects that more than half of global population growth between 2020 and 2050 will occur there.

Canada is the second largest country in the world by total area, spanning about 9.98 million square kilometres, and it has the longest coastline of any nation, measuring roughly 202,080 kilometres. Despite its size, Canada has a population of only about 40 million people, most of whom live within 200 kilometres of the southern border with the United States. The country contains more lakes than the rest of the world combined, an estimated 62 percent of the global total.

Chapter 4: Science and the Natural World

Photosynthesis is the process by which green plants, algae and some bacteria convert light energy, carbon dioxide and water into glucose and oxygen. It takes place in the chloroplasts, where the pigment chlorophyll absorbs light most strongly in the blue and red parts of the spectrum while reflecting green wavelengths. The overall reaction consumes six molecules of carbon dioxide and six molecules of water to produce one molecule of glucose and six molecules of oxygen, and it is the primary source of the oxygen in Earth's atmosphere.

Water is the only common substance on Earth that occurs naturally as a solid, a liquid and a gas within the range of surface temperatures. Its boiling point at standard atmospheric pressure is 100 degrees Celsius and its freezing point is 0 degrees Celsius; both values shift with pressure, which is why water boils at about 71 degrees Celsius on the summit of Mount Everest. Water has an unusually high specific heat capacity, allowing oceans to buffer the climate of coastal regions.

The speed of light in a vacuum is exactly 299,792,458 metres per second, a value now used to define the metre itself. Light from the Sun takes about eight minutes and twenty seconds to reach Earth, while the nearest star beyond the Sun, Proxima Centauri, is 4.24 light-years away. Because nothing can travel faster than light, astronomical distances are commonly expressed in light-years, the distance light covers in one year, approximately 9.46 trillion kilometres.

Chapter 5: Technology and the Modern World

The World Wide Web was invented in 1989 by the British computer scientist Tim Berners-Lee while he was working at CERN in Switzerland. He proposed a system of hypertext documents linked together and accessible over the internet, and he wrote the first web browser and server in 1990. In 1993 CERN released the underlying technology royalty-free, a decision widely credited with enabling the rapid global growth of the web; Berners-Lee has said the decision was essential to its success.

The Linux kernel was created in 1991 by Linus Torvalds, then a student at the University of Helsinki, and released under the GNU General Public License. Linux now powers the majority of web servers, all of the world's top 500 supercomputers and the Android operating system used by most smartphones. Its development model, in which thousands of contributors around the world review each other's code, is one of the most studied examples of large-scale open-source collaboration.

PostgreSQL is an open-source relational database management system with more than thirty-five years of active development, first released as POSTGRES at the University of California, Berkeley in 1986. It supports ACID transactions, extensible data types and a rich query language, and its extension ecosystem includes PostGIS for geospatial data and pgvector for storing and searching vector embeddings. The pgvector extension adds a vector data type and supports both exact and approximate nearest-neighbour search using distance operators.

Chapter 6: Retrieval-Augmented Generation

Retrieval-augmented generation, usually abbreviated RAG, is a technique that combines an information retrieval system with a language model. Instead of relying only on knowledge stored in its parameters, a RAG system first retrieves passages that are relevant to a user's question from an external corpus and then conditions the generated answer on those passages. This grounding reduces hallucination and allows the system to cite the exact source of each claim.

A typical RAG pipeline has two phases. In the ingestion phase, documents are parsed, split into overlapping chunks, converted into embedding vectors and written to a vector store. In the query phase, the user's question is embedded with the same model, the nearest chunks are retrieved by cosine similarity, optional reranking or hybrid search fuses lexical and semantic signals, and the top passages are passed to the answer stage together with their source metadata.

Hybrid retrieval combines dense vector similarity with sparse lexical matching such as BM25 or PostgreSQL full-text search. Because the two channels produce scores on different scales, systems often fuse them by rank rather than by score. Reciprocal rank fusion is a simple and robust method: each document receives the sum of one divided by a constant plus its rank in each list, so a document that appears near the top of both lists is promoted even when the raw scores are not comparable.

Chunking strategy strongly influences retrieval quality. Windows that are too small lose context and cannot answer questions whose evidence spans several sentences, while windows that are too large dilute the embedding and waste context space in the answer stage. A common compromise is a window of roughly a thousand characters with ten to twenty percent overlap, so that sentences crossing a boundary still appear intact in at least one chunk.